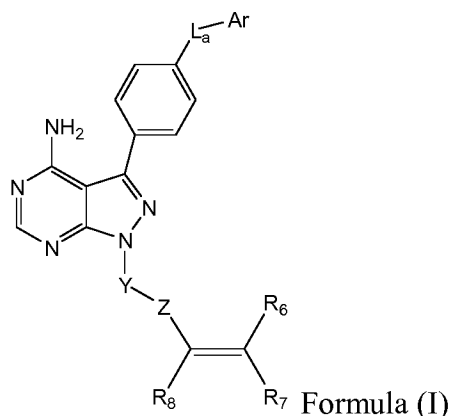


**WHAT IS CLAIMED IS:**

1. A method for treating a disorder characterized by the presence or development of one or more solid tumors comprising administering to an individual in need thereof a composition containing a therapeutically effective amount of a compound that forms a covalent bond with a cysteine sidechain of a Bruton's tyrosine kinase or a Bruton's tyrosine kinase homolog.
2. The method of claim 1, wherein the compound is a compound of Formula (I) having the structure:



wherein:

$L_a$  is  $CH_2$ , O, NH or S;

Ar is a substituted or unsubstituted aryl, or a substituted or unsubstituted heteroaryl; and either

(i) Y is an optionally substituted group selected from among alkylene,

heteroalkylene, arylene, heteroarylene, alkylenearylene, alkyleneheteroarylene, alkylencycloalkylene and alkyleneheterocycloalkylene;

Z is  $C(=O)$ ,  $NHC(=O)$ ,  $NR^aC(=O)$ ,  $NR^aS(=O)_x$ , where x is 1 or 2, and  $R^a$  is H, substituted or unsubstituted alkyl, or substituted or unsubstituted cycloalkyl; and

(a)  $R_7$  and  $R_8$  are H;

$R_6$  is H, substituted or unsubstituted  $C_1$ - $C_4$ alkyl, substituted or unsubstituted  $C_1$ - $C_4$ heteroalkyl,  $C_1$ - $C_8$ alkylaminoalkyl,  $C_1$ - $C_8$ hydroxyalkylaminoalkyl,  $C_1$ - $C_8$ alkoxyalkylaminoalkyl, substituted or unsubstituted  $C_3$ - $C_6$ cycloalkyl, substituted or unsubstituted  $C_1$ - $C_8$ alkyl $C_3$ - $C_6$ cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted  $C_2$ - $C_8$ heterocycloalkyl, substituted or unsubstituted

heteroaryl, C<sub>1</sub>-C<sub>4</sub>alkyl(aryl), C<sub>1</sub>-C<sub>4</sub>alkyl(heteroaryl), C<sub>1</sub>-C<sub>8</sub>alkylethers, C<sub>1</sub>-C<sub>8</sub>alkylamides, or C<sub>1</sub>-C<sub>4</sub>alkyl(C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl); or

(b) R<sub>6</sub> and R<sub>8</sub> are H;

R<sub>7</sub> is H, substituted or unsubstituted C<sub>1</sub>-C<sub>4</sub>alkyl, substituted or unsubstituted C<sub>1</sub>-C<sub>4</sub>heteroalkyl, C<sub>1</sub>-C<sub>8</sub>alkylaminoalkyl, C<sub>1</sub>-C<sub>8</sub>hydroxyalkylaminoalkyl, C<sub>1</sub>-C<sub>8</sub>alkoxyalkylaminoalkyl, substituted or unsubstituted C<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted C<sub>1</sub>-C<sub>8</sub>alkylC<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl, substituted or unsubstituted heteroaryl, C<sub>1</sub>-C<sub>4</sub>alkyl(aryl), C<sub>1</sub>-C<sub>4</sub>alkyl(heteroaryl), C<sub>1</sub>-C<sub>8</sub>alkylethers, C<sub>1</sub>-C<sub>8</sub>alkylamides, or C<sub>1</sub>-C<sub>4</sub>alkyl(C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl); or

(c) R<sub>7</sub> and R<sub>8</sub> taken together form a bond;

R<sub>6</sub> is H, substituted or unsubstituted C<sub>1</sub>-C<sub>4</sub>alkyl, substituted or unsubstituted C<sub>1</sub>-C<sub>4</sub>heteroalkyl, C<sub>1</sub>-C<sub>8</sub>alkylaminoalkyl, C<sub>1</sub>-C<sub>8</sub>hydroxyalkylaminoalkyl, C<sub>1</sub>-C<sub>8</sub>alkoxyalkylaminoalkyl, substituted or unsubstituted C<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted C<sub>1</sub>-C<sub>8</sub>alkylC<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl, substituted or unsubstituted heteroaryl, C<sub>1</sub>-C<sub>4</sub>alkyl(aryl), C<sub>1</sub>-C<sub>4</sub>alkyl(heteroaryl), C<sub>1</sub>-C<sub>8</sub>alkylethers, C<sub>1</sub>-C<sub>8</sub>alkylamides, or C<sub>1</sub>-C<sub>4</sub>alkyl(C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl); or

(ii) Y is an optionally substituted group selected from cycloalkylene or heterocycloalkylene;

Z is C(=O), NHC(=O), NR<sup>a</sup>C(=O), NR<sup>a</sup>S(=O)<sub>x</sub>, where x is 1 or 2, and R<sup>a</sup> is H, substituted or unsubstituted alkyl, substituted or unsubstituted cycloalkyl; and either

(a) R<sub>7</sub> and R<sub>8</sub> are H;

R<sub>6</sub> is substituted or unsubstituted C<sub>1</sub>-C<sub>4</sub>heteroalkyl, C<sub>1</sub>-C<sub>8</sub>hydroxyalkylaminoalkyl, C<sub>1</sub>-C<sub>8</sub>alkoxyalkylaminoalkyl, substituted or unsubstituted C<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted C<sub>1</sub>-C<sub>8</sub>alkylC<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl, substituted or unsubstituted heteroaryl, C<sub>1</sub>-C<sub>4</sub>alkyl(aryl), C<sub>1</sub>-C<sub>4</sub>alkyl(heteroaryl), C<sub>1</sub>-C<sub>8</sub>alkylethers, C<sub>1</sub>-C<sub>8</sub>alkylamides, or C<sub>1</sub>-C<sub>4</sub>alkyl(C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl);

(b) R<sub>6</sub> and R<sub>8</sub> are H;

R<sub>7</sub> is substituted or unsubstituted C<sub>1</sub>-C<sub>4</sub>heteroalkyl, C<sub>1</sub>-C<sub>8</sub>hydroxyalkylaminoalkyl, C<sub>1</sub>-C<sub>8</sub>alkoxyalkylaminoalkyl, substituted or unsubstituted C<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted C<sub>1</sub>-C<sub>8</sub>alkylC<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl, substituted or unsubstituted heteroaryl, C<sub>1</sub>-C<sub>4</sub>alkyl(aryl), C<sub>1</sub>-C<sub>4</sub>alkyl(heteroaryl), C<sub>1</sub>-C<sub>8</sub>alkylethers, C<sub>1</sub>-C<sub>8</sub>alkylamides, or C<sub>1</sub>-C<sub>4</sub>alkyl(C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl); or

(c) R<sub>7</sub> and R<sub>8</sub> taken together form a bond;

R<sub>6</sub> is substituted or unsubstituted C<sub>1</sub>-C<sub>4</sub>alkyl, substituted or unsubstituted C<sub>1</sub>-C<sub>4</sub>heteroalkyl, C<sub>1</sub>-C<sub>8</sub>alkylaminoalkyl, C<sub>1</sub>-C<sub>8</sub>hydroxyalkylaminoalkyl, C<sub>1</sub>-C<sub>8</sub>alkoxyalkylaminoalkyl, substituted or unsubstituted C<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted C<sub>1</sub>-C<sub>8</sub>alkylC<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl, substituted or unsubstituted heteroaryl, C<sub>1</sub>-C<sub>4</sub>alkyl(aryl), C<sub>1</sub>-C<sub>4</sub>alkyl(heteroaryl), C<sub>1</sub>-C<sub>8</sub>alkylethers, C<sub>1</sub>-C<sub>8</sub>alkylamides, or C<sub>1</sub>-C<sub>4</sub>alkyl(C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl); or a

pharmaceutically active metabolite, pharmaceutically acceptable solvate, pharmaceutically acceptable salt, or a pharmaceutically acceptable prodrug thereof.

3. The method of claim 2, wherein:

L<sub>a</sub> is O;

Ar is phenyl;

Z is C(=O), NHC(=O), or NCH<sub>3</sub>C(=O).

4. The method of claim 3, wherein

Y is alkyleneheterocycloalkylene;

R<sub>6</sub> and R<sub>8</sub> are H; and

R<sub>7</sub> is H, substituted or unsubstituted C<sub>1</sub>-C<sub>4</sub>alkyl, substituted or unsubstituted C<sub>1</sub>-C<sub>4</sub>heteroalkyl, C<sub>1</sub>-C<sub>8</sub>alkylaminoalkyl, C<sub>1</sub>-C<sub>8</sub>hydroxyalkylaminoalkyl, C<sub>1</sub>-C<sub>8</sub>alkoxyalkylaminoalkyl, substituted or unsubstituted C<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted C<sub>1</sub>-C<sub>8</sub>alkylC<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl, substituted or unsubstituted heteroaryl, C<sub>1</sub>-C<sub>4</sub>alkyl(aryl), C<sub>1</sub>-C<sub>4</sub>alkyl(heteroaryl), C<sub>1</sub>-C<sub>8</sub>alkylether, C<sub>1</sub>-C<sub>8</sub>alkylamide, or C<sub>1</sub>-C<sub>4</sub>alkyl(C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl).

5. The method of claim 3, wherein:

Y is an optionally substituted group selected from cycloalkylene or heterocycloalkylene; and

R<sub>6</sub> and R<sub>8</sub> are H; and

R<sub>7</sub> is substituted or unsubstituted C<sub>1</sub>-C<sub>4</sub>heteroalkyl, C<sub>1</sub>-C<sub>8</sub>hydroxyalkylaminoalkyl, C<sub>1</sub>-C<sub>8</sub>alkoxyalkylaminoalkyl, substituted or unsubstituted C<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted C<sub>1</sub>-C<sub>8</sub>alkylC<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl, substituted or unsubstituted heteroaryl, C<sub>1</sub>-C<sub>4</sub>alkyl(aryl), C<sub>1</sub>-C<sub>4</sub>alkyl(heteroaryl), C<sub>1</sub>-C<sub>8</sub>alkylether, C<sub>1</sub>-C<sub>8</sub>alkylamide, or C<sub>1</sub>-C<sub>4</sub>alkyl(C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl).

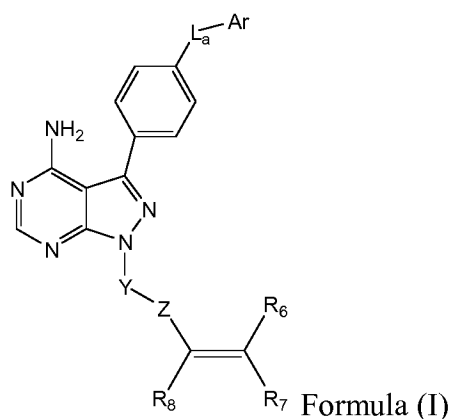
6. The method of claim 1, wherein the disorder is a sarcoma, lymphoma, or carcinoma.

7. The method of claim 1, wherein the disorder is mammary ductal carcinoma, lobular carcinoma, an adenocarcinoma, small cell lung carcinoma, non-small cell lung carcinoma, a melanoma, or a combination thereof.

8. The method of claim 1, wherein the disorder is pancreatic cancer.

9. A method for treating a disorder characterized by the presence or development of one or more solid tumors comprising administering to an individual in need thereof a composition containing a therapeutically effective amount of a compound that forms a covalent bond with a cysteine sidechain of a HER4.

10. The method of claim 9, wherein the compound is a compound of Formula (I) having the structure:



wherein:

L<sub>a</sub> is CH<sub>2</sub>, O, NH or S;

Ar is a substituted or unsubstituted aryl, or a substituted or unsubstituted heteroaryl; and either

(i) Y is an optionally substituted group selected from among alkylene, heteroalkylene, arylene, heteroarylene, alkylenearylene, alkyleneheteroarylene, alkylene cycloalkylene and alkyleneheterocycloalkylene;

Z is C(=O), NHC(=O), NR<sup>a</sup>C(=O), NR<sup>a</sup>S(=O)<sub>x</sub>, where x is 1 or 2, and R<sup>a</sup> is H, substituted or unsubstituted alkyl, substituted or unsubstituted cycloalkyl; and either

(a) R<sub>7</sub> and R<sub>8</sub> are H;

R<sub>6</sub> is H, substituted or unsubstituted C<sub>1</sub>-C<sub>4</sub>alkyl, substituted or unsubstituted C<sub>1</sub>-C<sub>4</sub>heteroalkyl, C<sub>1</sub>-C<sub>8</sub>alkylaminoalkyl, C<sub>1</sub>-C<sub>8</sub>hydroxyalkylaminoalkyl, C<sub>1</sub>-C<sub>8</sub>alkoxyalkylaminoalkyl, substituted or unsubstituted C<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted C<sub>1</sub>-C<sub>8</sub>alkylC<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl, substituted or unsubstituted heteroaryl, C<sub>1</sub>-C<sub>4</sub>alkyl(aryl), C<sub>1</sub>-C<sub>4</sub>alkyl(heteroaryl), C<sub>1</sub>-C<sub>8</sub>alkylether, C<sub>1</sub>-C<sub>8</sub>alkylamide, or C<sub>1</sub>-C<sub>4</sub>alkyl(C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl);

(b) R<sub>6</sub> and R<sub>8</sub> are H;

R<sub>7</sub> is H, substituted or unsubstituted C<sub>1</sub>-C<sub>4</sub>alkyl, substituted or unsubstituted C<sub>1</sub>-C<sub>4</sub>heteroalkyl, C<sub>1</sub>-C<sub>8</sub>alkylaminoalkyl, C<sub>1</sub>-C<sub>8</sub>hydroxyalkylaminoalkyl, C<sub>1</sub>-C<sub>8</sub>alkoxyalkylaminoalkyl, substituted or unsubstituted C<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted C<sub>1</sub>-C<sub>8</sub>alkylC<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl, substituted or unsubstituted heteroaryl, C<sub>1</sub>-C<sub>4</sub>alkyl(aryl), C<sub>1</sub>-C<sub>4</sub>alkyl(heteroaryl), C<sub>1</sub>-C<sub>8</sub>alkylether, C<sub>1</sub>-C<sub>8</sub>alkylamide, or C<sub>1</sub>-C<sub>4</sub>alkyl(C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl); or

(c) R<sub>7</sub> and R<sub>8</sub> taken together form a bond;

R<sub>6</sub> is H, substituted or unsubstituted C<sub>1</sub>-C<sub>4</sub>alkyl, substituted or unsubstituted C<sub>1</sub>-C<sub>4</sub>heteroalkyl, C<sub>1</sub>-C<sub>8</sub>alkylaminoalkyl, C<sub>1</sub>-C<sub>8</sub>hydroxyalkylaminoalkyl, C<sub>1</sub>-C<sub>8</sub>alkoxyalkylaminoalkyl, substituted or unsubstituted C<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted C<sub>1</sub>-C<sub>8</sub>alkylC<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl, substituted or unsubstituted heteroaryl, C<sub>1</sub>-C<sub>4</sub>alkyl(aryl), C<sub>1</sub>-C<sub>4</sub>alkyl(heteroaryl), C<sub>1</sub>-C<sub>8</sub>alkylether, C<sub>1</sub>-C<sub>8</sub>alkylamide, or C<sub>1</sub>-C<sub>4</sub>alkyl(C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl); or

(ii) Y is an optionally substituted group selected from cycloalkylene or heterocycloalkylene;

Z is C(=O), NHC(=O), NR<sup>a</sup>C(=O), NR<sup>a</sup>S(=O)<sub>x</sub>, where x is 1 or 2, and R<sup>a</sup> is H, substituted or unsubstituted alkyl, substituted or unsubstituted cycloalkyl; and either

(a) R<sub>7</sub> and R<sub>8</sub> are H;

R<sub>6</sub> is substituted or unsubstituted C<sub>1</sub>-C<sub>4</sub>heteroalkyl, C<sub>1</sub>-C<sub>8</sub>hydroxyalkylaminoalkyl, C<sub>1</sub>-C<sub>8</sub>alkoxyalkylaminoalkyl, substituted or unsubstituted C<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted C<sub>1</sub>-C<sub>8</sub>alkylC<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl, substituted or unsubstituted heteroaryl, C<sub>1</sub>-C<sub>4</sub>alkyl(aryl), C<sub>1</sub>-C<sub>4</sub>alkyl(heteroaryl), C<sub>1</sub>-C<sub>8</sub>alkylethers, C<sub>1</sub>-C<sub>8</sub>alkylamides, or C<sub>1</sub>-C<sub>4</sub>alkyl(C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl);

(b) R<sub>6</sub> and R<sub>8</sub> are H;

R<sub>7</sub> is substituted or unsubstituted C<sub>1</sub>-C<sub>4</sub>heteroalkyl, C<sub>1</sub>-C<sub>8</sub>hydroxyalkylaminoalkyl, C<sub>1</sub>-C<sub>8</sub>alkoxyalkylaminoalkyl, substituted or unsubstituted C<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted C<sub>1</sub>-C<sub>8</sub>alkylC<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl, substituted or unsubstituted heteroaryl, C<sub>1</sub>-C<sub>4</sub>alkyl(aryl), C<sub>1</sub>-C<sub>4</sub>alkyl(heteroaryl), C<sub>1</sub>-C<sub>8</sub>alkylethers, C<sub>1</sub>-C<sub>8</sub>alkylamides, or C<sub>1</sub>-C<sub>4</sub>alkyl(C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl); or

(c) R<sub>7</sub> and R<sub>8</sub> taken together form a bond;

R<sub>6</sub> is substituted or unsubstituted C<sub>1</sub>-C<sub>4</sub>alkyl, substituted or unsubstituted C<sub>1</sub>-C<sub>4</sub>heteroalkyl, C<sub>1</sub>-C<sub>8</sub>alkylaminoalkyl, C<sub>1</sub>-C<sub>8</sub>hydroxyalkylaminoalkyl, C<sub>1</sub>-C<sub>8</sub>alkoxyalkylaminoalkyl, substituted or unsubstituted C<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted C<sub>1</sub>-C<sub>8</sub>alkylC<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl, substituted or unsubstituted heteroaryl, C<sub>1</sub>-C<sub>4</sub>alkyl(aryl), C<sub>1</sub>-C<sub>4</sub>alkyl(heteroaryl), C<sub>1</sub>-C<sub>8</sub>alkylethers, C<sub>1</sub>-C<sub>8</sub>alkylamides, or C<sub>1</sub>-C<sub>4</sub>alkyl(C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl); and

pharmaceutically active metabolites, or pharmaceutically acceptable solvates, pharmaceutically acceptable salts, or pharmaceutically acceptable prodrugs thereof.

11. The method of claim 10, wherein:  
L<sub>a</sub> is O;  
Ar is phenyl;  
Z is C(=O), NHC(=O), or NCH<sub>3</sub>C(=O).
- 5 12. The method of claim 11, wherein  
Y is alkyleneheterocycloalkylene;  
R<sub>6</sub> and R<sub>8</sub> are H; and  
R<sub>7</sub> is H, substituted or unsubstituted C<sub>1</sub>-C<sub>4</sub>alkyl, substituted or unsubstituted C<sub>1</sub>-C<sub>4</sub>heteroalkyl, C<sub>1</sub>-C<sub>8</sub>alkylaminoalkyl, C<sub>1</sub>-C<sub>8</sub>hydroxyalkylaminoalkyl, C<sub>1</sub>-C<sub>8</sub>alkoxyalkylaminoalkyl, substituted or unsubstituted C<sub>3</sub>-C<sub>6</sub>cycloalkyl,  
10 C<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted C<sub>1</sub>-C<sub>8</sub>alkylC<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl, substituted or unsubstituted heteroaryl, C<sub>1</sub>-C<sub>4</sub>alkyl(aryl), C<sub>1</sub>-C<sub>4</sub>alkyl(heteroaryl), C<sub>1</sub>-C<sub>8</sub>alkylether, C<sub>1</sub>-C<sub>8</sub>alkylamide, or C<sub>1</sub>-C<sub>4</sub>alkyl(C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl).
- 15 13. The method of claim 11, wherein:  
Y is an optionally substituted group selected from cycloalkylene or heterocycloalkylene; and  
R<sub>6</sub> and R<sub>8</sub> are H; and  
R<sub>7</sub> is substituted or unsubstituted C<sub>1</sub>-C<sub>4</sub>heteroalkyl, C<sub>1</sub>-C<sub>8</sub>hydroxyalkylaminoalkyl,  
20 C<sub>1</sub>-C<sub>8</sub>alkoxyalkylaminoalkyl, substituted or unsubstituted C<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted C<sub>1</sub>-C<sub>8</sub>alkylC<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl, substituted or unsubstituted heteroaryl, C<sub>1</sub>-C<sub>4</sub>alkyl(aryl), C<sub>1</sub>-C<sub>4</sub>alkyl(heteroaryl), C<sub>1</sub>-C<sub>8</sub>alkylethers, C<sub>1</sub>-C<sub>8</sub>alkylamides, or C<sub>1</sub>-C<sub>4</sub>alkyl(C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl).
- 25 14. The method of claim 9, wherein the disorder is mammary ductal carcinoma, lobular carcinoma, or a combination thereof.
15. A method for treating a disorder characterized by the presence or development of one or more solid tumors comprising administering to an individual in need thereof a kinase inhibitor that selectively and irreversibly binds to a protein tyrosine kinase selected from  
30 Btk, a Btk homolog, an ACK, HER4, and a Btk kinase cysteine homolog, in which the kinase inhibitor reversibly and non-selectively binds to a multiplicity of protein tyrosine kinases, and further in which the plasma half life of the kinase inhibitor is less than about 4 hours.

16. The method of claim 15, wherein the kinase inhibitor selectively and irreversibly binds to Btk.

17. The method of claim 15, wherein the kinase inhibitor irreversibly binds to HER4.

18. The method of claim 15, wherein the disorder is a sarcoma, lymphoma, and/or carcinoma.

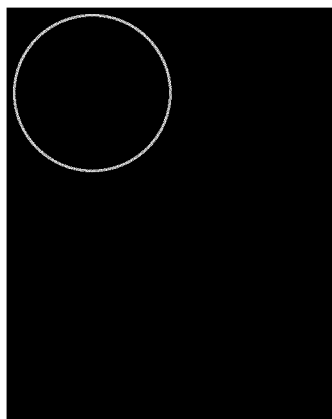
19. The method of claim 15, wherein the disorder is mammary ductal carcinoma, lobular carcinoma, an adenocarcinoma, small cell lung carcinoma, non-small cell lung carcinoma, a melanoma, or a combination thereof.

20. The method of claim 15, wherein the disorder is mammary ductal carcinoma, lobular carcinoma, or a combination thereof.

21. The method of claim 15, wherein the disorder is pancreatic cancer.

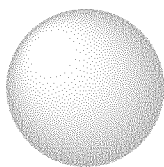
22. The method of claim 15, in which the plasma half life of the kinase inhibitor is less than about 3 hours.

23. The method of claim 15, wherein the kinase inhibitor has the structure of Formula (VII):



Formula (VII)

wherein:



is a moiety that binds to the active site of a kinase, including a tyrosine kinase, further including a Btk kinase cysteine homolog;

Y is an optionally substituted group selected from among alkylene, heteroalkylene, arylene, heteroarylene, heterocycloalkylene, cycloalkylene, alkylenearylene, alkyleneheteroarylene, alkylencycloalkylene, and alkyleneheterocycloalkylene; Z is C(=O), OC(=O), NHC(=O), NCH<sub>3</sub>C(=O), C(=S), S(=O)<sub>x</sub>, OS(=O)<sub>x</sub>, NHS(=O)<sub>x</sub>, where x is 1 or 2;

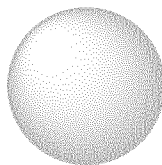


R<sub>6</sub> and R<sub>8</sub> are independently selected from among H, unsubstituted C<sub>1</sub>-C<sub>4</sub> alkyl, substituted C<sub>1</sub>-C<sub>4</sub>alkyl, unsubstituted C<sub>1</sub>-C<sub>4</sub>heteroalkyl, substituted C<sub>1</sub>-C<sub>4</sub>heteroalkyl, unsubstituted C<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted C<sub>3</sub>-C<sub>6</sub>cycloalkyl, unsubstituted C<sub>2</sub>-C<sub>6</sub>heterocycloalkyl, and substituted C<sub>2</sub>-C<sub>6</sub>heterocycloalkyl; or

R<sub>6</sub> and R<sub>8</sub> taken together form a bond;

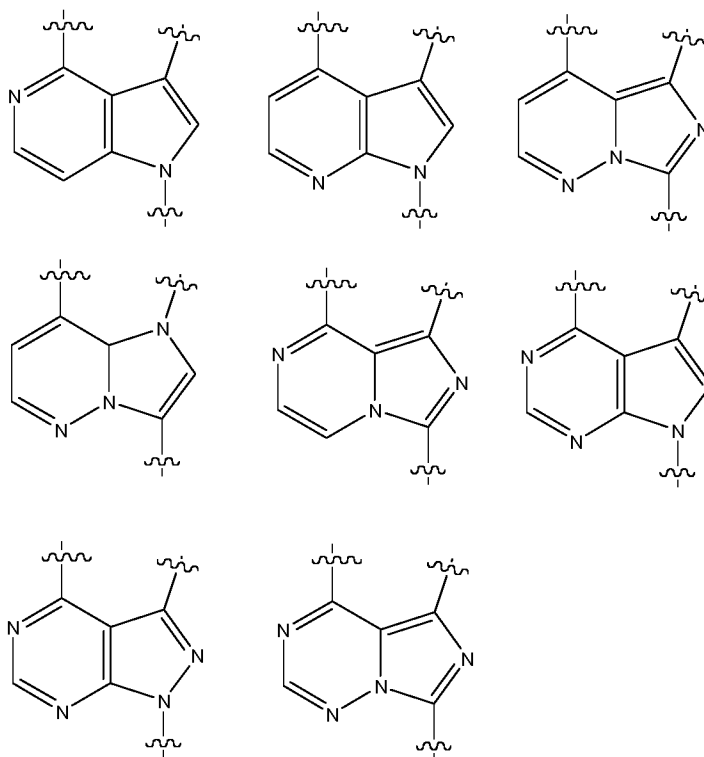
R<sub>7</sub> is H, substituted or unsubstituted C<sub>1</sub>-C<sub>4</sub>alkyl, substituted or unsubstituted C<sub>1</sub>-C<sub>4</sub>heteroalkyl, C<sub>1</sub>-C<sub>6</sub>alkoxyalkyl, C<sub>1</sub>-C<sub>8</sub>alkylaminoalkyl, C<sub>1</sub>-C<sub>8</sub>hydroxyalkylaminoalkyl, substituted or unsubstituted C<sub>3</sub>-C<sub>6</sub>cycloalkyl, substituted or unsubstituted aryl, substituted or unsubstituted C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl, substituted or unsubstituted heteroaryl, C<sub>1</sub>-C<sub>4</sub>alkyl(aryl), C<sub>1</sub>-C<sub>4</sub>alkyl(heteroaryl), C<sub>1</sub>-C<sub>4</sub>alkyl(C<sub>3</sub>-C<sub>8</sub>cycloalkyl), or C<sub>1</sub>-C<sub>4</sub>alkyl(C<sub>2</sub>-C<sub>8</sub>heterocycloalkyl); or a

pharmaceutically active metabolite, or pharmaceutically acceptable solvate, pharmaceutically acceptable salt, or pharmaceutically acceptable prodrug thereof.



24. The method of claim 21, wherein selected from

is a substituted fused biaryl moiety



25. The method of claim 22, wherein:

Z is C(=O), NHC(=O), NCH<sub>3</sub>C(=O), or S(=O)<sub>2</sub>;

each of R<sub>6</sub> and R<sub>8</sub> is H; or

R<sub>6</sub> and R<sub>8</sub> taken together form a bond.

26. The method of claim 23, wherein:

Y is a 4-, 5-, 6-, or 7-membered cycloalkylene ring; or

Y is a 4-, 5-, 6-, or 7-membered heterocycloalkylene ring; or

Y is a C<sub>1</sub>-C<sub>4</sub> alkylene, or 4-, 5-, 6-, or 7-membered heterocycloalkylene ring;

R<sub>7</sub> is H, substituted or unsubstituted C<sub>1</sub>-C<sub>4</sub>alkyl, substituted or unsubstituted C<sub>1</sub>-

C<sub>4</sub>heteroalkyl, C<sub>1</sub>-C<sub>6</sub>alkoxyalkyl, C<sub>1</sub>-C<sub>8</sub>alkylaminoalkyl, C<sub>1</sub>-

C<sub>8</sub>hydroxyalkylaminoalkyl, C<sub>1</sub>-C<sub>8</sub>alkoxyalkylaminoalkyl, C<sub>1</sub>-C<sub>4</sub>alkyl(aryl), C<sub>1</sub>-

C<sub>4</sub>alkyl(heteroaryl), C<sub>1</sub>-C<sub>4</sub>alkyl(C<sub>3</sub>-C<sub>8</sub>cycloalkyl), or C<sub>1</sub>-C<sub>4</sub>alkyl(C<sub>2</sub>-

C<sub>8</sub>heterocycloalkyl).